cozaint®

MARCIA™ EBOOK · SECOND EDITION

The Architecture of Intelligent Video Lifecycle Management

Why enterprise surveillance needs more than disk



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About this edition

This book began in 2023 as *Video Surveillance Storage: Implementing a 2-Tier Storage Solution* and was revised through January 2026 (revision 2.08). Its argument has not changed: surveillance video should be recorded at the quality the cameras were bought for, kept for as long as it is needed, and stored on the technology that fits each stage of its life, not crammed onto ever-larger racks of spinning disk.

What has changed is how practical that argument has become. The first edition described the ideal: an operator who simply moves the video management system (VMS) timeline and gets back any recording, whether it sits on disk or on LTO tape, with no extra steps and no IT staff. Cozaint has since built that workflow into **MARCIA™ Intelligent Video Lifecycle Management** and patented it (U.S. Patent No. 12,755,902).

This second edition keeps the educational core of the first: why disk-only storage becomes expensive, what LTO is, how a multi-tier architecture works, how to size it and what it costs. It adds what we have learned since: why retained video is becoming more valuable, how the patented workflow works, what the economics look like at enterprise scale, how Hollywood's experience points the way, how ATTO XstreamCORE bridges share an LTO library over Ethernet, and how MARCIA supports an unbreakable chain of custody for video evidence.

The idea in one sentence. Media & Entertainment proved LTO for video; MARCIA makes it practical for surveillance; patented VMS integration makes the storage tier invisible to the operator. The result is substantially more retained video at dramatically lower lifecycle cost.

How this book is organized

Part	Chapters
The problem	1. The video storage problem · 2. Why video is worth keeping
The technology	3. LTO: not your grandfather's tape · 4. Why disk alone falls short
The architecture	5. The architecture of Intelligent Video Lifecycle Management · 6. MARCIA and the patented workflow · 7. Hollywood comes to video surveillance
The decision	8. What about the cloud? · 9. Sizing a system · 10. The economics · 11. Real-world examples

Part	Chapters
The implementation	12. LTO library physical connection to the server · 13. Advanced connectivity: shared LTO via Ethernet with ATTO XstreamCORE · 14. The unbreakable chain of custody · 15. Best practices and getting started

Foreword to the first edition

Written for the 2023 edition and reproduced as published.

Welcome to this introduction of what I believe is a useful ebook on 2-tier video surveillance storage systems utilizing Linear Tape-Open (LTO) technology as the second tier. I'm Rich Gadomski, Head of Tape Evangelism at FUJIFILM, and I'm thrilled to be sharing with you some insights about this fascinating and increasingly relevant topic.

As you know, video surveillance has become an essential tool in many sectors of our society, from law enforcement to retail, transportation, and critical infrastructure. The ability to capture and analyze video footage has enabled organizations to enhance their security, prevent crimes, optimize operations, and make data-driven decisions. However, with the proliferation of high-definition cameras, the exponential growth of video data, and the increasing demand for longer retention periods, managing video surveillance storage has become a significant challenge for many IT and security departments.

That's where the concept of 2-tier video surveillance storage systems comes in. Rather than storing all video data on expensive, energy intensive, high-performance storage devices such as hard disk drives, organizations can leverage a 2-tiered approach that provides for the quick access of the most recent video and the availability of all recorded video no matter when it was originally stored. This approach can significantly reduce storage costs, optimize system performance, reduce carbon footprint and simplify video data management.

But what is LTO technology, and why is it a useful component of a 2-tier video surveillance storage system? Linear Tape-Open, or LTO, is an open-format magnetic tape technology that was first introduced in 2000 by IBM, HPE, and Quantum. Since then, LTO has become a widely adopted standard for data backup and archiving, thanks to its high capacity, fast transfer rates, reliability, low energy consumption and low cost per gigabyte. In a 2-tier video surveillance storage system, LTO serves as the second tier where older or less frequently accessed video data is replicated from the primary storage layer, such as hard disk arrays.

By using LTO tapes, organizations can store massive amounts of video data in a cost-effective and scalable manner, while also ensuring data integrity and long-term retention. LTO tapes can currently provide up to 18TB of capacity per

cartridge and transfer rates of up to 400MB/s, making them ideal for large-scale video surveillance environments. And these capacities and speeds are already set to increase.

But LTO technology is not just useful for its cost, capacity, energy profile and speed. LTO also offers advanced features that are critical for video surveillance storage, such as data encryption, WORM (Write-Once-Read-Many) functionality, and LTFS (Linear Tape File System) support. With data encryption, organizations can protect their video data from unauthorized access or theft, while WORM functionality ensures that once data is written to the tape, it cannot be overwritten or erased, thus preserving the authenticity and legal admissibility of the data. LTFS enables users to access, manage, and share video data on LTO tapes as if they were on a disk or a network drive, simplifying data access and reducing playback times.

In today's video surveillance infrastructure, where data volumes are growing exponentially, and regulatory and compliance requirements are becoming more stringent, 2-tier video surveillance storage systems utilizing LTO technology as the second tier are becoming very compelling. This book will delve deeper into the technical aspects of 2-tier video surveillance storage systems utilizing LTO technology, covering topics such as system architecture, data replication, LTO versus hard disk only offerings, LTFS and more. It will also provide practical tips and best practices for designing, implementing, and managing such systems, based on real-world scenarios and use cases.

Rich Gadomski, Head of Tape Evangelism, FUJIFILM Recording Media, USA, Inc. (2023)

1. The video storage problem

Video surveillance has become an essential part of modern security. High-quality recordings deter crime, speed up investigations, support prosecutions, protect workers and help organizations run better. Video management software (VMS) records, stores and manages that video, and for most of the industry's history the VMS has written everything to hard disk.

That approach is becoming more expensive every year, for reasons that compound one another:

- **More cameras.** Organizations keep adding them, and every camera adds video around the clock.
- **Higher resolution and frame rates.** A single 4K camera can generate up to 12 GB of video per hour. Higher quality is the reason cameras are bought, and it multiplies storage.
- **Longer retention.** Regulations, policies, litigation and investigations all push retention from days to months or years.
- **The cost of disk does not stop at purchase.** Hard drives spin and draw power 24 hours a day, every watt must be cooled, drives fail and wear out, and larger arrays need more enclosures, servers, rack space and support.

The industry has been solving the wrong end of the problem

Faced with those costs, the surveillance industry has mostly changed the video instead of the storage:

- lowering camera **resolution**, which discards detail needed to identify people and objects;
- lowering **frame rate**, which loses fast motion;
- lowering **bit rate**, which degrades quality;
- **recording only on motion**, which misses events that the motion settings did not catch;
- and, when all else fails, **shortening retention**, which simply throws evidence away.

Each of these saves disk at the expense of the very thing the cameras were bought to capture. Organizations purchase high-definition cameras to obtain the best video available, and then record that video degraded or not at all.

> **The purpose of this book** is to remove those misguided methods of dealing with the video storage problem: record all video at the best quality, so as not to compromise safety, security or business value, and change the storage architecture instead.

Why "single-tier" thinking persists

Most surveillance systems treat storage as one tier: whatever the VMS records goes to disk and stays there until it is overwritten. Other industries that manage large volumes of data stopped thinking this way long ago. They place data on the storage that matches how often it is used: fast, expensive storage for active data; dense, low-cost storage for everything else. Surveillance video, which is written once and rarely viewed after the first few days, is an ideal candidate for the same approach.

2. Why video is worth keeping

The first edition focused on cost: storage is expensive, and a multi-tier architecture makes it affordable. That is still true. But there is a bigger reason to rethink surveillance storage: **historical video is becoming more valuable.**

Storage savings start the conversation. Preserving Trusted Visual Evidence is why it matters.

The clip that matters is often the one nobody knew mattered

Reason to keep video	What happens without it
Investigations	Fraud, theft and misconduct are often discovered weeks or months after the fact
Litigation and liability	Slip-and-fall claims, disputes and lawsuits are frequently filed long after the event; without video, organizations cannot defend themselves
Compliance	Records rules, gaming regulations, corrections policies and evidence rules set minimum retention periods
Incident reconstruction	Understanding how an event unfolded often requires video from before anyone knew there was an event
AI re-analysis	New analytics can re-examine history for patterns, people, vehicles and events, but only if the history still exists

Quality matters as much as quantity

Retained video is only useful if it is good enough to use. The ability to identify a face, read a plate or see a card switched at a blackjack table depends on resolution and frame rate. Video kept at reduced quality to save disk may be present but useless.

Chain of custody

Video submitted as evidence must be the original, unaltered recording, handled in a way that can be documented. An archive that keeps the original recordings and returns them through the same VMS used to record them supports that chain of custody far better than ad hoc copies and exports. Chapter 14 covers the legal challenges video evidence faces and how MARCIA's tracking of LTO cartridge data strengthens the chain of custody.

If video is an asset, the question is no longer “how little can we keep?” but “how can we afford to keep what we need, at full quality, and still get it back quickly?” The rest of this book answers that question.

3. LTO: not your grandfather's tape

Linear Tape-Open (LTO) is a high-capacity, reliable, energy-efficient and cost-effective digital data storage technology. It was introduced in 2000, is developed by Hewlett Packard Enterprise, IBM and Quantum through the LTO Program, and has become the industry standard for backup and archiving. It is also heavily used in Hollywood and broadcasting to archive video.

This is not VHS

When people in security hear "tape," many think of the VHS cassettes once used in analog surveillance systems. The two have almost nothing in common:

	VHS (analog video tape)	LTO (digital data tape)
Purpose	Consumer analog video recording	Enterprise digital data storage and archiving
Capacity	Up to about 6 hours of analog video	30 TB per LTO-10 cartridge, up to 40 TB with the newest cartridges
Quality over time	Degrades with age and each playback	Digital; error correction verifies data as it is written
Recording	Helical scan	Linear serpentine recording at up to hundreds of MB per second
Lifespan	Years	Media manufacturers rate LTO for up to 50 years of archival life

Why LTO fits surveillance video

Video is a **sequential** stream: each frame follows the one before it. LTO writes and reads data sequentially, one block after another, as the tape moves past the read/write head. That makes surveillance video and LTO a natural match: video is written to tape in the same order it was recorded and can be read back the same way.

Surveillance video also has the profile LTO was designed for:

- **vast amounts of data;**
- **rarely accessed after the first few days,** until it is needed;
- **sequential** data;
- a desire for **long-term retention.**

The advantages of LTO

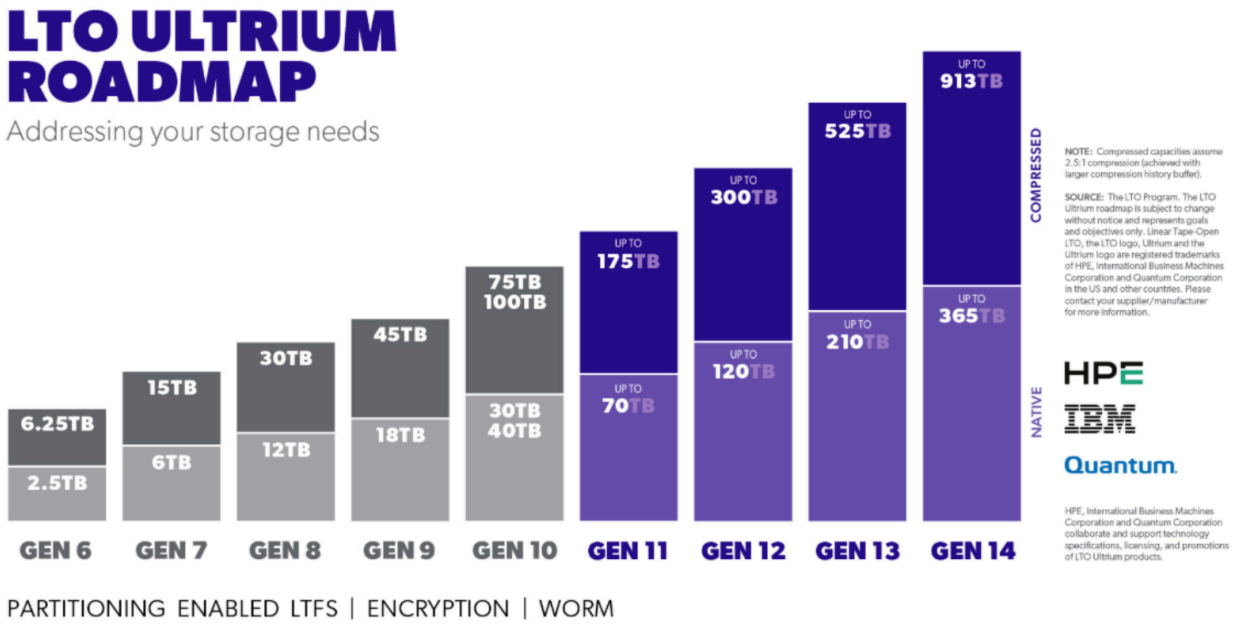
- **Scalability.** LTO has a published roadmap of generations with predictable capacity growth, extending to LTO-14 at a planned 576 TB per cartridge, more than half a petabyte on one palm-sized cartridge.
- **Reliability.** Built-in error correction and verification; modern LTO bit error rates are orders of magnitude better than hard disk.
- **Cost.** A very low cost per terabyte and a low total cost of ownership.
- **Durability.** Rugged cartridges protect the tape from dust and handling, and manufacturers rate the media for up to 50 years.
- **Energy.** A cartridge draws power only while it is being read or written. In a library slot or on a shelf it draws none and needs no cooling. Studies cited by the tape industry show tape systems using about 87% less energy than an equivalent capacity of hard disk over time, and producing about 97% less CO2e over their lifecycle.
- **Security.** Hardware encryption, write-once (WORM) cartridges, and the option to keep cartridges offline, out of reach of the network.

LTFS: tape that describes itself

The Linear Tape File System (LTFS), introduced in 2010, is an open standard that lets a tape be read much like a removable drive. One partition holds the content and the other holds its index, so the cartridge is self-describing. Users can browse a tape, find files and read them directly without restoring the whole tape, on many operating systems and without proprietary software. For long-term archives, that openness matters: the archive is not locked to one server or one vendor.

As Carnegie Mellon professor James Bain put it: “The message about tape is that, just as SSDs did not displace HDDs, HDDs, in turn will not displace tape systems. All of them co-exist in a tiered storage architecture where cost and latency are balanced at every level of the system.”

The LTO generations



The LTO Ultrium roadmap: native and compressed capacity by generation, from LTO-6 to LTO-14 (source: the LTO Program)

Generation	Year	Native capacity	Notable addition
LTO-1	2000	100 GB	
LTO-2	2003	200 GB	
LTO-3	2005	400 GB	WORM
LTO-4	2007	800 GB	Drive-level encryption
LTO-5	2010	1.5 TB	LTFS
LTO-6	2012	2.5 TB	Barium ferrite media
LTO-7	2015	6 TB	
LTO-8	2017	12 TB	
LTO-9	2021	18 TB	Up to 400 MB/s
LTO-10	2025	30 TB (up to 40 TB with the newest cartridges)	
LTO-11 to LTO-14	Roadmap	Up to a planned 576 TB	

Tape libraries

An LTO library (sometimes called a “jukebox”) holds cartridges in slots and one or more tape drives, with a robot that moves cartridges between them. The library keeps track of where every cartridge is, so data can be retrieved without anyone touching a tape.

Libraries scale from a compact 2U unit with a single drive and a couple of dozen slots to enterprise libraries with dozens of drives and thousands of cartridges, holding many petabytes. Most grow by adding expansion modules, drives and cartridges, so a system can start small and grow with retention.

4. Why disk alone falls short

Hard disk drives (HDDs) are the right place for video that is being recorded and reviewed now. They are the wrong place to keep months or years of video that is rarely viewed.

The costs that keep coming

- **Capacity costs scale with retention.** Double the retention and you double the disk.
- **Power and cooling.** Disks spin 24 hours a day, and every watt they draw must also be cooled.
- **Failures and wear-out.** Drives are mechanical. They fail at random during service and wear out after about five years of continuous use, so long-lived archives must replace every drive at least once.
- **RAID overhead, space and support.** RAID 6 typically uses two of every twelve drives for protection, and more drives mean more enclosures, servers, rack space and support contracts.

The hard disk roadmap

Hard disk capacity has grown enormously since IBM's first 5 MB drive in 1956, and heat- and microwave-assisted recording (HAMR, MAMR) push it further. But as drives approach the physical limits of areal density, capacity growth is slower and costlier than on the LTO roadmap, and every additional terabyte still spins and draws power.

Shingled magnetic recording (SMR) drives raise capacity by overlapping tracks, but they are a poor fit for surveillance recording: writes are slower, RAID rebuilds are difficult and reads can bottleneck. SMR drives should be avoided in video surveillance recording systems.

What about solid-state drives?

SSDs offer fast access and low latency, but for large-scale video retention they are the most expensive option per terabyte, have a finite number of write cycles that continuous recording consumes, and still draw power whenever they are online. They have a place in high-performance tiers, not in the long-term archive.

The work-arounds, revisited

Because disk is costly, the industry has tried to make video smaller. None of these addresses the real issue, that disk is the wrong medium for long-term retention:

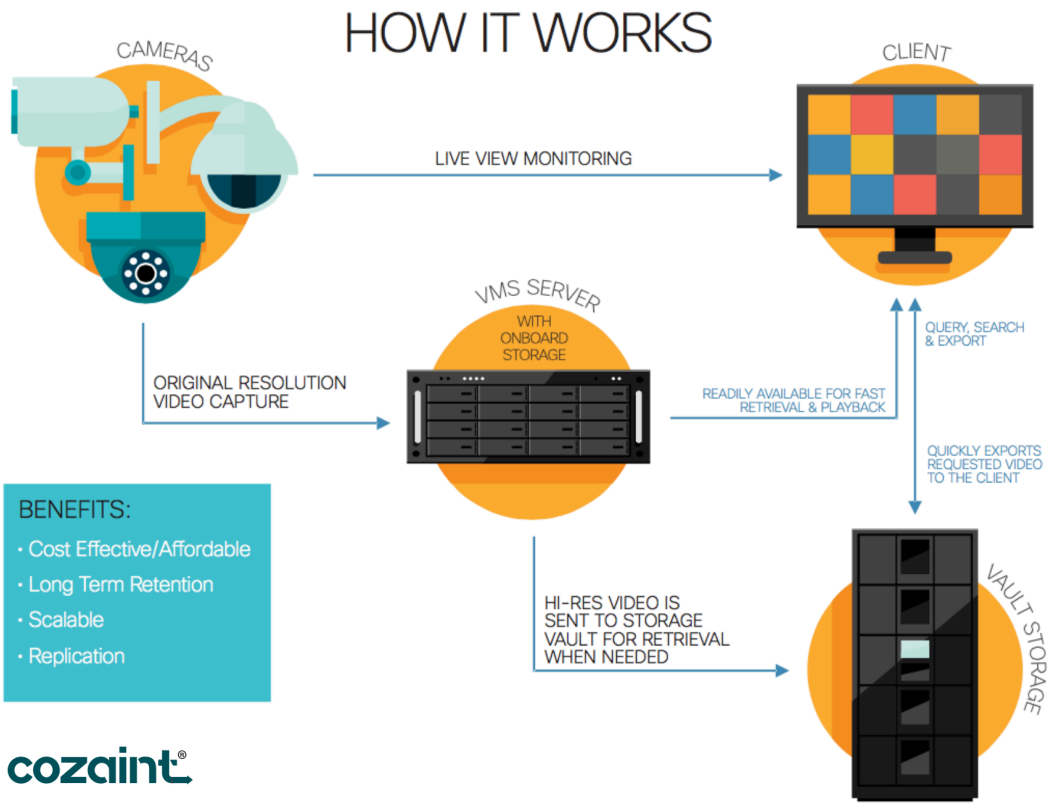
- **Compression (H.264, H.265, H.265+)** reduces storage substantially and is good practice. But as camera counts and resolutions grow, the need for more storage returns, and the cycle starts again.
- **Record-on-motion** stores video only "where there is activity." Trees, birds and flickering lights trigger false recordings, so thresholds are added, and then systems fail to record when something important happens.
- **Lower resolution, frame rate and bit rate** degrade the evidence.

5. The architecture of Intelligent Video Lifecycle Management

A multi-tier architecture puts each stage of a recording's life on the storage that fits it. The first edition called this "2-tier video surveillance storage." Today Cozaint calls the complete approach **Intelligent Video Lifecycle Management (IVLM)**: not just two kinds of storage, but a governed lifecycle that keeps the operator in control from the moment video is recorded to the moment it is retrieved years later.

Three stages, one connected lifecycle

Stage	What happens	Where the video lives
01 Active video	The VMS records and plays recent video as it always has	A right-sized disk tier
02 Governed retention	Retention rules decide what moves to LTO and when; new video is replicated to tape automatically in the background	LTO cartridges in the library, and offline if exported
03 Connected workflow	Requested history returns to the operator through a defined workflow	Recalled from LTO to the same VMS timeline



How a two-tier system works: cameras record to the VMS server's disk tier, video is replicated to the LTO vault, and the operator's client queries, plays back and exports any recording

Tier 1: disk for active video

The first tier holds the most recent video, typically days to a few weeks, on high-performance disk. This is where operators review most incidents, so playback must be instant. Because it holds only a fraction of the total video, it can be a fraction of the size of a disk-only system. A common starting point is about 10% of the total retention, adjusted to how far back operators usually look.

Tier 2: LTO for retention

The second tier holds **every** recording for the full retention period on LTO cartridges in a tape library. It is the long-term record: very low cost per terabyte, no power for cartridges at rest, decades of archive life, and the option of offline copies.

Because video is replicated to tape as it is recorded, the archive is not a backup made after the fact. It grows continuously, and the disk tier can simply roll over recent video once its copy is safely on tape.

The VMS must understand the tiers

Here is the part most earlier attempts missed. It is not enough to attach a tape library to a recording server. The VMS, or the layer that works with it, must understand where every piece of video lives, so that:

- data moves from disk to tape according to policy, without operator effort;
- the system knows, for any camera and any moment, which tier holds the video and, on tape, which cartridge and position;
- an operator's request for older video can be answered automatically;
- the recalled video returns to the VMS for playback and export.

Historically, VMS vendors had little incentive to support LTO. Hard disk NAS and SAN storage were universal, and end customers did not yet know to ask for a tiered solution. That is the gap MARCIA fills.

The timeline is the key

Every VMS gives operators a **timeline**: a horizontal view of recorded time for each camera. Operators move along it to the moment they need, play, pause, scrub forward and backward, and export clips for HR, police or the courts.

In an ideal multi-tier system, nothing about that changes. The operator moves the timeline to the date and time of interest:

- if the video is on disk, it plays immediately;
- if it is on LTO, the system identifies the cartridge, the library loads it into a drive, the drive seeks to the right point, and the video comes back to the timeline, typically within a few minutes;
- either way, there are no extra steps and no IT staff involved.

Properly sizing the disk tier, and setting expectations with operators, delivers an infrastructure in which any recording is available either instantly or within minutes. Most operators readily accept that, and organizations realize substantial savings over disk-only storage.

The editor and the operator

Video editors in film and television also work on a timeline, in a non-linear editor (NLE). The two are close cousins, with two important differences:

	Editor's timeline	Operator's timeline
Length	Finite: one production	Endless: every camera, every day
Does the user know what they need?	Yes, the editor knows the shot	Often not; the operator is searching

Those differences explain why earlier ways of using tape failed in surveillance:

- **Pre-fetch** assumes the operator knows in advance what to restore. They usually don't; that is why they are reviewing video.
- **Backup and restore** makes tape an IT process. Where is the video restored to, and which system plays it? Every step and every ticket makes the archive less likely to be used.

The lesson: **security is not an IT process. The surveillance operator is king.** The archive must come to the operator's timeline.

Two ways to connect a VMS and an archive

Integration between a VMS and LTO can be built through a middleware layer that translates VMS requests into storage commands, or through an application programming interface (API) that lets the two systems communicate directly. Either way, the measure of success is the same: whether the operator can ignore where the video is stored.

6. MARCIA and the patented workflow

Cozaint MARCIA™ Intelligent Video Lifecycle Management is the layer between the VMS and long-term storage that makes the architecture in Chapter 5 work in practice. Its patented architecture lets an operator's normal VMS playback request recall archived video from LTO and play it on the same timeline they use every day.

U.S. Patent No. 12,755,902, issued October 6, 2026 to Cozaint Corporation (application No. 18/065,506).

The five steps

Request.
The operator moves the VMS timeline to a past date and time.

Locate.
MARCIA identifies which storage tier holds that video.

Map.
It maps the request to the exact cartridge and position.

Recall.
The tape library loads the cartridge and reads the video.

Play.
The video returns to the same VMS timeline.

Step	What the operator sees	What MARCIA does
1 Request	Moves the timeline, exactly as for recent video	Receives the playback request for that camera and time
2 Locate	Nothing new	Determines whether the video is on disk or on LTO
3 Map	Nothing new	Maps camera and time to the cartridge and position where the video was written
4 Recall	A visible retrieval status	Directs the library to load the cartridge and reads the requested video
5 Play	The video plays on the same timeline	Returns the original recording to the VMS for playback and export

Why this is different

Competitors can sell tape libraries. Competitors can sell hierarchical storage management (HSM) software. Competitors can build disk archives. None of those, on their own, makes LTO part of the operator's normal workflow. The patent explains why MARCIA is not simply another tape product: it is the integration that makes the storage tier invisible to the operator.

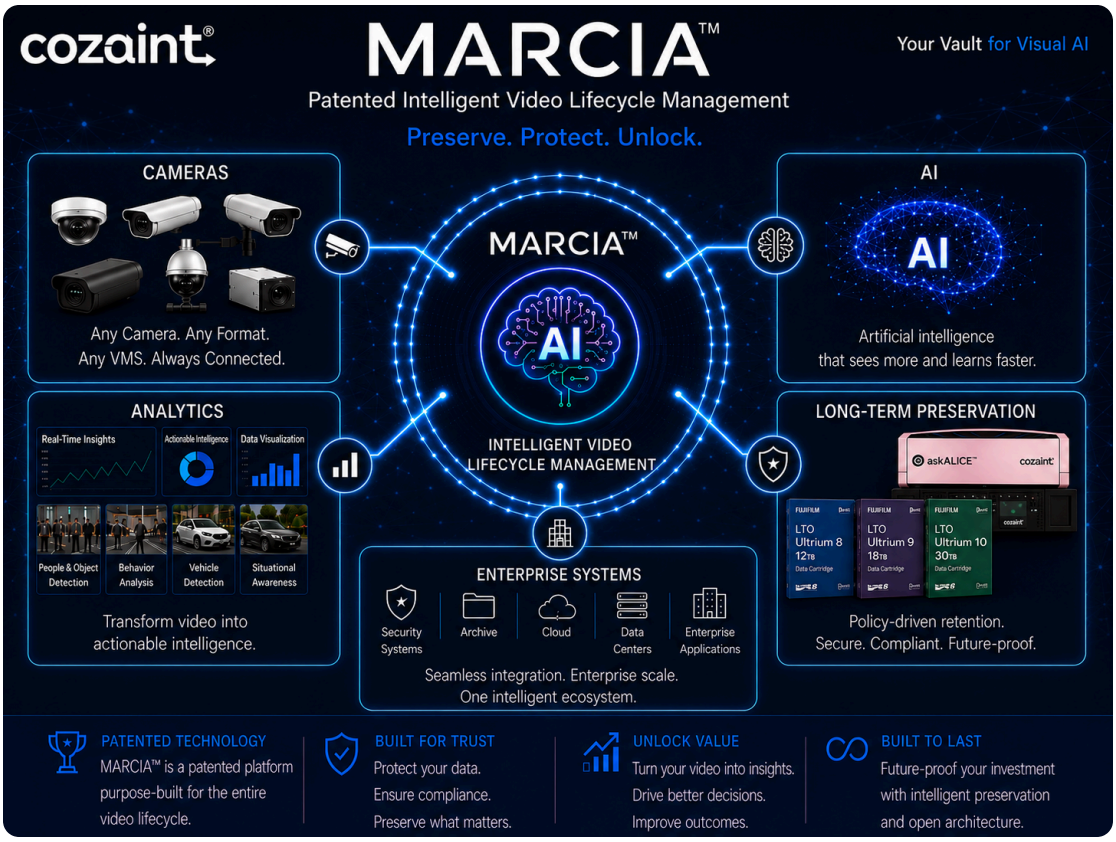
Design choices that matter

Choice	Why it matters
Video for each camera kept together on tape	Recalling a camera's history touches as few cartridges as possible
At least two LTO drives	One drive can serve a recall while another keeps writing new video
Policy-driven movement	Retention follows the organization's rules
Visible request and retrieval status	Operators know their video is on the way
Open LTO format (LTFS)	Cartridges are self-describing and readable on standard systems
Offline cartridges and WORM media	Air-gapped copies and write-once recordings where policy requires them

> **Being precise about protection.** A cartridge is air-gapped only when it is removed from the library or otherwise disconnected; cartridges in the library are online. Recordings are protected from being overwritten when written to WORM media or governed by retention policy. Each deployment should be designed to the level of protection the organization's policy requires.

Compatibility

MARCIA works with Milestone XProtect, Hanwha WAVE, Digital Watchdog DW Spectrum, Network Optix Nx Witness and Cozaint BOBBYvms, and is validated on HPE StoreEver and Quantum Scalar tape libraries and ATTO XstreamCORE.



MARCIA: Intelligent Video Lifecycle Management connects cameras, analytics, AI and long-term preservation

One platform, three ways to deploy

	MARCIA™	askALICE™	ALICE1™
What it is	The Intelligent Video Lifecycle Management platform	Enterprise turnkey disk-plus-LTO system, powered by MARCIA	Compact LTO archive appliance, powered by MARCIA
Best for	Enterprises adding a retention tier to the VMS they run	Large sites that want one complete system	Smaller and self-contained sites



Cozaint ALICE1: a single LTO drive and a four-bay hard disk video surveillance system in one compact appliance, powered by MARCIA

ALICE1 brings the same architecture to smaller and self-contained sites: recent video on its built-in disk cache, long-term retention on LTO cartridges loaded in its own drive, and playback from the VMS timeline. askALICE pairs an enterprise recording server with a full LTO library (Chapter 12), and with ATTO XstreamCORE several askALICE systems can share one library over Ethernet (Chapter 13).

7. Hollywood comes to video surveillance

Dailies. B-roll. Circle takes. For more than a decade, the film and television production business has lived with an explosion of recorded video, and has built entirely new workflows to handle it.

In the days of film, parts of movies, TV shows and commercials ended up on the cutting-room floor. Many in Media & Entertainment (M&E) point to digital cinema cameras such as RED, first conceived in 2005, as the tipping point. Once the cost of film stock disappeared, productions began to keep everything: every take might be reused in a bloopers reel, a director's cut or a sequel. One small side effect: there is a lot of extra video.

The questions Hollywood already answered

How do directors and producers quickly review today's shoot? How do editors find a scene recorded for another production years ago and reuse it? What is the most affordable way to store all of it? Hollywood has answered each of these, and the surveillance industry faces the same questions with the same kind of data.

Two things Hollywood got right

Metadata. Editors tag footage so it can be found later. "Airplane flyover of the Golden Gate Bridge at sunset" is a clip a production company can find in seconds and reuse rather than sending another crew to shoot it. In surveillance, video analytics plays the same role: people, vehicles, objects and events make years of footage searchable, and that metadata becomes worth its weight in gold.

Multiple tiers of storage. Just as in surveillance, production video is rarely viewed again after the first days. Hollywood keeps active footage on fast disk and archives everything to LTO. LTO meets the needs of both the studios and the insurance and bonding companies behind many productions, which have required footage captured on set or on location to be archived to LTO daily, on the same tape-based platform that banks use.

The lesson for surveillance

Hollywood also learned that if you cannot get back to recorded video quickly and easily, there is little point in keeping it. Just like an editor, a surveillance operator cannot afford to wait or to hand the search to someone else. Operators need to scrub forward and backward through the timeline without concern for where the video is stored, then select the clip they need, directing the system to the original, untouched recording on its LTO cartridge. Done this way, multi-tier storage also delivers the chain of custody needed to submit video as evidence.

From a cost standpoint, the comparison is stark. Compare the acquisition cost of terabytes and petabytes of spinning NAS and SAN storage with an LTO library complete with robotics. Add the three-to-five-year life of spinning disks against decades of LTO archive life. Add the electricity and cooling for all that spinning disk, 24/7, year-round.

When contemplating a 21st-century surveillance video management solution, a disk-only approach is like navigating before anyone knew the world was round. There are better ways to reach our destination. We all want to store as much video as we can, at the highest resolution and the highest frame rates, as affordably as possible. So why not learn from our colleagues in Hollywood? Embracing the sequential structure of video and LTO data tape delivers the ideal storage solution, and done properly, LTO multi-tier storage is surveillance video's "killer app."

Thanks, Hollywood.

8. What about the cloud?

Cloud storage is often presented as the easy answer for video retention: flexible, scalable and cheap to start. For long-term surveillance retention, the costs and limits are rarely discussed up front.

Bandwidth

Every camera's video must be uploaded continuously. A 1080p camera at 30 frames per second can need on the order of 20 Mbps, and a 4K camera at 60 frames per second around 100 Mbps. Sites with many high-definition cameras need dedicated high-speed connections, often fiber, which carry significant monthly costs and installation fees, and even a 1 Gbps connection may not be enough. Any outage in connectivity affects the recording system.

Recurring storage fees

Cloud providers charge every month for every terabyte stored. The calculator behind Chapter 10 uses the average standard, pay-as-you-go price of four leading providers, about $16.50 to $17 per TB per month at archive scale. Keeping one petabyte for a year at that price is roughly $200,000, and the bill never stops while the video is kept.

Fees to get your video back

Many providers charge egress or retrieval fees to download data, along with request charges, minimum storage periods and early-deletion fees. Investigations that need to pull large amounts of video can trigger significant, unpredictable bills.

Ownership, control and security

Organizations must check who controls their data, how it can be accessed, and what rights they retain to access, download and delete it. Video is sensitive, and moving it off-site means relying on another organization's security practices, often with less visibility into who accessed what.

And still the workflow problem

Even when the costs are acceptable, cloud archives often require operators to use a separate portal rather than the VMS timeline they use every day.

	Disk only	Cloud	MARCIA (disk + LTO)
Immediate access to recent video	Yes	Yes	Yes, recent tier on disk
Multi-year economics	Poor	Variable, recurring fees	Strong

	Disk only	Cloud	MARCIA (disk + LTO)
Power for archived video	Continuous	Provider's	None for cartridges at rest
Depends on internet bandwidth	No	Yes	No
Offline (air-gapped) copy	Limited	No	Yes, cartridges can be stored offline
Operator workflow	VMS timeline	Often a separate portal	Same VMS timeline
Scaling to petabytes	Expensive	Expensive, recurring	Designed for it

9. Sizing a system

Sizing a lifecycle architecture starts with the same question as any surveillance storage design: how much video will be recorded, and for how long must it be kept? The difference is what happens next. Instead of buying disk for the whole answer, you size a small disk tier for active video and an LTO tier for the full retention period.

What determines the amount of video

Factor	Effect
Number of cameras	More cameras, more video
Resolution	The single biggest factor: more pixels per frame, more data
Frame rate	More frames per second, more data
Motion	Busy scenes compress less than quiet ones
Compression	H.264 and H.265 reduce the bit rate substantially
Retention	How many days of video must be kept, which a lifecycle architecture lets you choose rather than settle for

Camera resolutions at a glance

Resolution is the primary element in the storage calculation: the higher the resolution, the more pixels in every frame, and the clearer and more detailed the image. High-resolution cameras are increasingly popular because they give greater detail and accuracy in identifying people and objects, and they need more storage.

Name	Resolution (pixels)	Aspect ratio	Megapixels	Pixels per second at 30 fps
QCIF	176 × 120	1.47:1	0.02	0.6 million
CIF	320 × 240	4:3	0.08	2.3 million
2CIF	704 × 240	2.93:1	0.17	5.1 million
4CIF	704 × 480	1.47:1	0.34	10.1 million
VGA	640 × 480	4:3	0.31	9.2 million
D1	720 × 480	3:2	0.35	10.4 million
WVGA	752 × 480	1.57:1	0.36	10.8 million

Name	Resolution (pixels)	Aspect ratio	Megapixels	Pixels per second at 30 fps
SVGA	800 × 600	4:3	0.48	14.4 million
XGA	1024 × 768	4:3	0.79	23.6 million
720p HD	1280 × 720	16:9	0.92	27.6 million
960p HD	1280 × 960	4:3	1.23	36.9 million
WXGA	1280 × 800	16:10	1.02	30.7 million
SXGA (1.3 MP)	1280 × 1024	5:4	1.31	39.3 million
HD+	1600 × 900	16:9	1.44	43.2 million
UXGA (2 MP)	1600 × 1200	4:3	1.92	57.6 million
1080p Full HD	1920 × 1080	16:9	2.07	62.2 million
2K (DCI)	2048 × 1080	1.90:1	2.21	66.4 million
QXGA (3 MP)	2048 × 1536	4:3	3.15	94.4 million
WQHD (1440p)	2560 × 1440	16:9	3.69	110.6 million
QSXGA (5.2 MP)	2560 × 2048	5:4	5.24	157.3 million
5 MP	2592 × 1944	4:3	5.04	151.2 million
3K	3072 × 1728	16:9	5.31	159.3 million
4K (DCI)	4096 × 2160	1.90:1	8.85	265.4 million
20 MP	5472 × 3648	3:2	19.96	598.9 million
8K	8192 × 4608	16:9	37.75	1,132.5 million

Pixels per second is the raw image data a camera captures before compression; compression (H.264, H.265) reduces what is actually stored by a large factor.

From bit rate to terabytes

The most practical way to size storage is from each camera's recording bit rate, the amount of data it records per second after compression:

- **Data per camera per day** = bit rate (kilobits per second) ÷ 8 × 86,400 seconds
- **Total storage** = data per camera per day × number of cameras × retention days

Cozaint's storage calculator at cozaint.com/assessment estimates the bit rate from resolution, frame rate and motion, or lets you enter it directly, and does the arithmetic.

Compression

H.264 (also called AVC) divides each frame into blocks and uses motion estimation, transform coding and entropy coding to remove redundancy between and within frames. It produces far smaller files than older standards such as MPEG-2 while keeping quality high. H.265 (HEVC) compresses further at the cost of more processing power. Motion JPEG compresses each frame separately and produces much larger files. Compression is good practice, but it does not change the underlying trend: as cameras and resolutions grow, storage grows with them.

Frame rate: why more is often better

The human eye does not see in frames, but studies suggest people can distinguish separate images at around 60 frames per second or more under some conditions. Higher frame rates capture motion with greater temporal detail, and when played back in slow motion they reveal what normal speed hides.

Casino surveillance is a good example. In games such as blackjack, slow-motion review helps surveillance teams spot card switching, chip manipulation, dealer collusion or improper shuffling. A player swapping a low card for a high one from a sleeve may be visible only frame by frame. The only reason to record at less than the camera's full frame rate is storage cost. Remove that constraint, and the argument favors recording at 30 frames per second or more.

A worked example

Input	Value
Cameras	50
Resolution	1080p
Average motion	30%
Frame rate	30 fps
Compression	H.264, about 3,110 kbps per camera
Retention	180 days
Total video	**about 281.5 TB**

Disk-only design. A traditional single-tier design needs enough RAID-protected disk for all 281.5 TB: for example, a 32-bay server populated with large drives in RAID.

Lifecycle design. Working with the operators, the disk tier is set at about 10% of the total, roughly 28 TB, rounded up to 30 TB to allow for RAID protection and overhead. That fits in a small four-bay NAS-class server. The LTO tier is a single 2U library with 24 slots, using only 16 LTO-9 cartridges for the full 180 days.

Comparing equipment and energy costs, the lifecycle design came to about **45% lower overall cost** than the traditional design, and it scales by adding cartridges as retention grows.

Tape drives and throughput

A single LTO drive can keep up with many concurrent camera streams. Best practice is at least two drives, so one can serve an operator's recall while recording continues to be written. Larger installations use larger libraries with more drives and hundreds of slots. The calculator estimates the drive count from the daily volume of video and the time needed to write each cartridge.

10. The economics

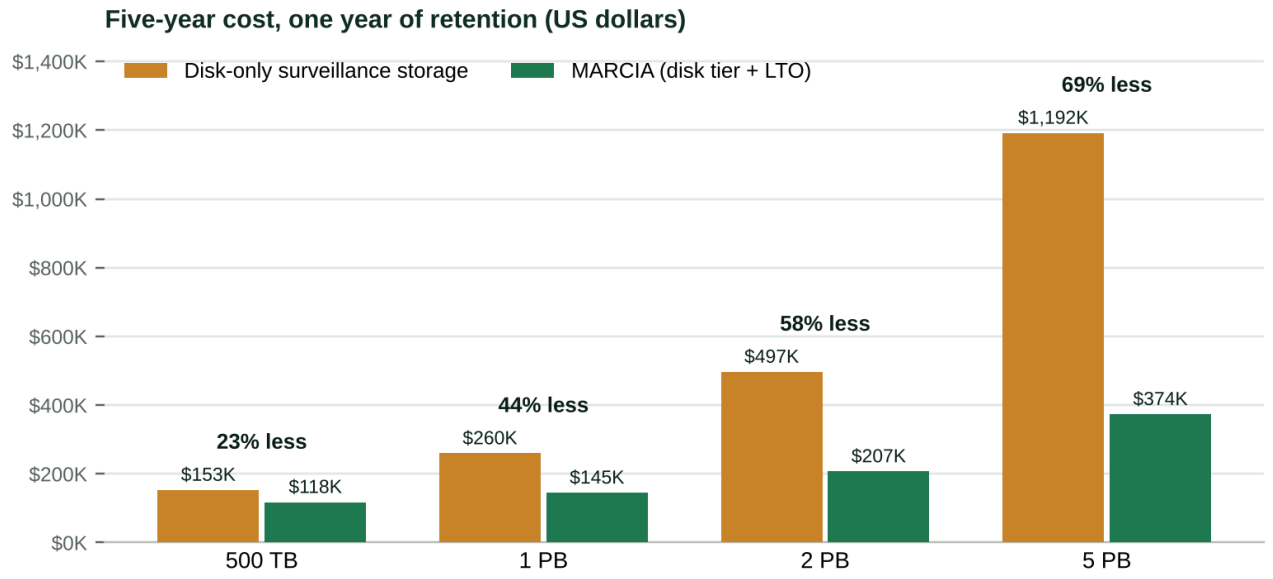
Most surveillance storage decisions are made on purchase price, but most of the money is spent afterwards: power and cooling around the clock, replacement drives, and above all the cost of keeping more video every year.

The companion paper *The Economics of Long-Term Video Retention* compares three ways to keep **one year** of video, using the same price model as the storage calculator at cozaint.com/assessment (prices checked November 1, 2026):

- **Disk-only:** the average price per terabyte of six complete video surveillance storage systems, with RAID 6, volume price breaks, power and cooling.
- **Cloud:** the average standard storage price of four cloud providers.
- **MARCIA:** a 10% disk tier plus LTO-9, with Cozaint products at full list price.

Five-year cost

Archive size	MARCIA (disk tier + LTO)	Disk-only system	Cloud storage	MARCIA saves vs disk	MARCIA saves vs cloud
500 TB	$117,762	$153,229	$508,680	23%	77%
1 PB	$145,062	$259,628	$1,003,080	44%	86%
2 PB	$207,050	$496,739	$1,991,880	58%	90%
5 PB	$373,530	$1,191,981	$4,958,280	69%	92%



Five-year cost of keeping one year of video: disk-only vs MARCIA

What the numbers show

1. **Scale changes the answer.** The tape tier carries fixed costs (a library, drives, software), so it pays off as the archive grows. Above 1 PB, MARCIA's advantage is large and keeps growing.
2. **Retention changes the answer.** Disk cost grows in direct proportion to the video kept. With MARCIA, keeping more video mostly means adding cartridges.
3. **Cloud is the most expensive way to keep a year of video,** before counting upload bandwidth and retrieval fees.
4. **The model leans in the competitors' favor.** Disk and cloud are at discounted or published prices, Cozaint at full list price, and the disk figures leave out rack space, support contracts and full system replacement after five years.

Across a range of earlier studies and deployments, organizations have typically realized **40% to 60% savings** over traditional surveillance storage with a two-tier disk-and-LTO design, and more as archives reach petabyte scale.

Energy and sustainability

LTO-based storage is a leading "green" choice for surveillance retention. Cartridges at rest draw no power and need no cooling, and studies cited by the tape industry put tape's long-term energy use about 87% below an equivalent capacity of disk, with about 97% less CO2e over the product lifecycle. That helps organizations meet ESG goals as well as budgets.

11. Real-world examples

The two-tier disk-and-LTO architecture has been applied in large and demanding surveillance environments.

New York City Police Department

The NYPD operates one of the largest video surveillance systems in the world, with more than 18,000 cameras deployed across the city. It implemented a two-tier storage solution with LTO technology to manage its video: an active tier of high-performance SSD and HDD storage for short-term retention and a cold tier of LTO tape for long-term retention. The NYPD estimated that the two-tier architecture reduced the cost of storage by **more than 70%** compared to traditional storage solutions.

NASA Kennedy Space Center

NASA's Kennedy Space Center implemented a two-tier video surveillance storage solution using LTO technology to manage the large amounts of video generated by its facilities. Short-term video is kept on high-performance storage, and long-term retention on LTO tapes, which are stored off-site for disaster recovery. The architecture enabled the center to store and manage its video efficiently and cost-effectively.

University of Oregon

The University of Oregon implemented a two-tier solution with LTO technology for its campus-wide surveillance system, with an active tier for short-term video and LTO for long-term retention. The university estimated that the architecture reduced the cost of storage by **more than 50%** compared to traditional storage solutions.

City of Oceanside, California

At the City of Oceanside's San Luis Rey Wastewater Treatment Plant, California records rules require certain government records, including video, to be kept for one year. More than 30 high-definition (1080p) cameras record 24/7, which comes to about 548 TB for a year. A disk-only system was too expensive for the City's budget.

The City deployed Cozaint's askALICE system, the disk-and-LTO architecture that MARCIA is built on. Using the FUJIFILM video surveillance retention cost calculator with the plant's own requirements, the two-tier design came to **$31,515 over five years against $68,506 for disk-only, a 54% saving.** Operators play back any recording from any point in the year themselves, with no help from IT.

> “We needed to upgrade our security surveillance system at our water treatment plant, and since we have worked with several different systems, we knew what we liked and what we didn’t like. It is always difficult to find a system that fits within our budget and meets our long-term retention requirements. Cozaint’s askALICE has been an ideal fit for our surveillance recording needs and our budget.”
>
> Scott Prather, Senior IT Analyst, City of Oceanside, California

The full story is in the case study *The Problem MARCIA Was Built to Solve*.

Return on investment

Organizations that adopt a lifecycle architecture can expect a return on their investment within a few years, and the savings grow as video accumulates. The scalability and longevity of LTO keep delivering value long after the purchase.

12. LTO library physical connection to the server

As discussed throughout this book, efficient storage is crucial in data-intensive environments such as video surveillance, and LTO offers the most efficient, reliable and cost-effective storage resources available. This chapter outlines how to physically connect an LTO library (some call these devices “jukeboxes”) to a PC server using a Serial Attached SCSI (SAS) host bus adapter (HBA), and the optional connection using a Fibre Channel (FC) HBA. It also explains how to connect more than one library to a single server with a hydra cable, and how the server recognizes and uses the connected library.



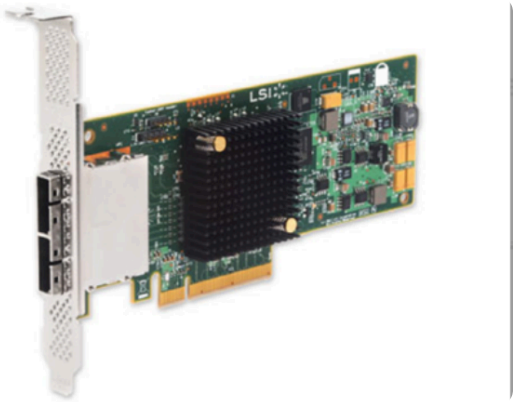
An LTO library with multiple LTO drives. One drive, or several, can support a large library with many cartridge slots.

Components and terminology

Component	What it is
LTO library	A device (sometimes called a “jukebox”) that houses one or more LTO tape drives and slots holding LTO cartridges for data storage and retrieval
PC server	The computer that runs the video management software and manages the LTO library and drives
Host bus adapter (HBA)	A card, typically PCIe, installed in the server that handles communication between the server and external storage devices

Component	What it is
Serial Attached SCSI (SAS)	A high-speed data transfer protocol used to connect storage devices
Fibre Channel (FC)	A high-speed network technology commonly used for storage connectivity

Connecting the LTO library to the server with a SAS HBA



Example of a SAS host bus adapter

Requirements. Before you begin, make sure you have:

- an LTO library with a SAS interface;
- a SAS HBA card;
- Mini-SAS cables;
- a PC server with an available PCIe slot;
- SAS drivers compatible with the server's operating system.

Instructions.

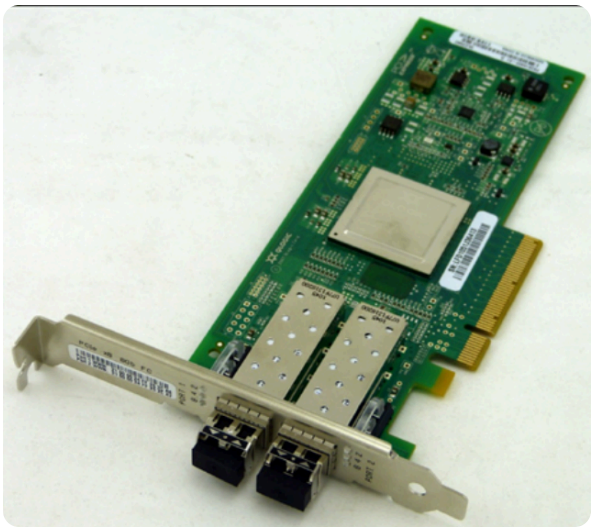
1. Power off the PC server and the LTO library.
2. Open the server chassis and locate an available PCIe slot.
3. Insert the SAS HBA card into the PCIe slot and secure it with the appropriate screws.
4. Connect one end of the Mini-SAS cable to the SAS port on the HBA and the other end to the SAS port on the LTO library. Make sure the connection is snug. The cable has keyed connectors to ensure correct alignment.
5. Power on the LTO library first and make sure it is completely ready.
6. Power on the PC server.
7. Install the SAS drivers for the server's operating system. These can usually be downloaded from the HBA manufacturer's website.

8. Once the drivers are installed, the HBA should detect the LTO library. Verify this in the Device Manager (Windows) or with the manufacturer's management software. You may need to reboot the server for the drivers to take effect.
9. Initialize and configure the LTO library according to the manufacturer's instructions.

Connecting the LTO library with a Fibre Channel HBA

If you want to add an FC HBA for higher performance and redundancy, you will need these additional components:

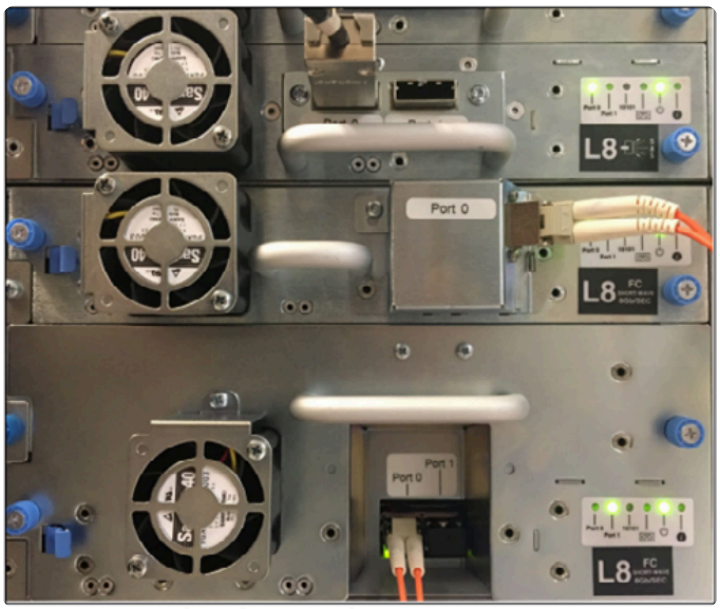
- an FC HBA card;
- FC cables;
- an FC switch, if you are connecting multiple devices.



Example of a Fibre Channel HBA

Instructions.

1. Power off the PC server and the LTO library.
2. Open the server chassis and locate an available PCIe slot.
3. Insert the FC HBA card into the PCIe slot and secure it with screws.
4. Connect one end of the FC cable to the FC HBA port on the server and the other end to the FC switch. If you are connecting directly to the LTO library, connect the cable directly to the library.
5. Power on the LTO library, the PC server and the FC switch (if used).
6. Install the FC drivers for the server's operating system, usually available from the HBA manufacturer's website.
7. Once the drivers are installed, the FC HBA should detect the FC network and the LTO library.
8. Configure the FC HBA settings, including zoning if you use an FC switch, according to the manufacturer's guidelines.



The back panel of an LTO library with Fibre Channel connectors

Connecting multiple LTO libraries to one server

Connecting several LTO libraries to a single server increases storage capacity and redundancy. Consider these factors first:

1. **Sufficient PCIe slots:** the server needs enough available slots for the additional HBAs.
2. **Resource allocation:** plan for the processing power, memory and bandwidth needed for multiple library connections.
3. **Operating system support:** check that the server's operating system supports multiple HBAs and libraries.
4. **Cable management:** organize cables to prevent tangling and keep airflow inside the server chassis.

Instructions.

1. Install SAS and/or FC HBAs for each LTO library you want to connect, following the steps above. Install each HBA in a separate PCIe slot and connect it to its library.
2. Install the required drivers for each HBA and configure them according to the manufacturer's guidelines.
3. Initialize and configure each LTO library individually, making sure each is recognized and accessible by the server.
4. Depending on the server's operating system and software, configure the system to use the multiple LTO libraries efficiently.

The hydra cable and how to use it

A hydra cable is a multi-connector cable that consolidates several SAS connections into a single cable. It simplifies cable management and improves airflow inside the server chassis. To use one:

1. Connect the individual SAS cables to the corresponding ports on the hydra cable.
2. Connect the other end of the hydra cable to the SAS HBA card.



A multi-connector “hydra” cable

Connecting an LTO library to a PC server with a SAS or FC HBA is a critical step in building a reliable video storage solution. Following these instructions gives a secure and efficient connection, and planning for multiple library connections lets the storage infrastructure scale with the demands of your video surveillance retention.

Built to scale



How LTO libraries stack and scale for greater retention: from a 3U base module with 3 drives and 40 cartridges to a 21U library with 21 drives and 280 cartridges

The same architecture runs from a compact 2U library with a single drive and two dozen slots, through stacked modular libraries and full 42U racks, to enterprise libraries with dozens or hundreds of drives and thousands of cartridges, such as the IBM TS4500, HPE StoreEver MSL6480, Quantum Scalar i6000 and Spectra Logic T-Finity families. Most grow by adding modules, drives and cartridges, so a system can start small and expand with retention.



The askALICE PC server (top) integrated with the LTO library (bottom) to provide a complete two-tier video surveillance storage solution

References and resources

For further information on LTO library connections, SAS HBAs and FC HBAs:

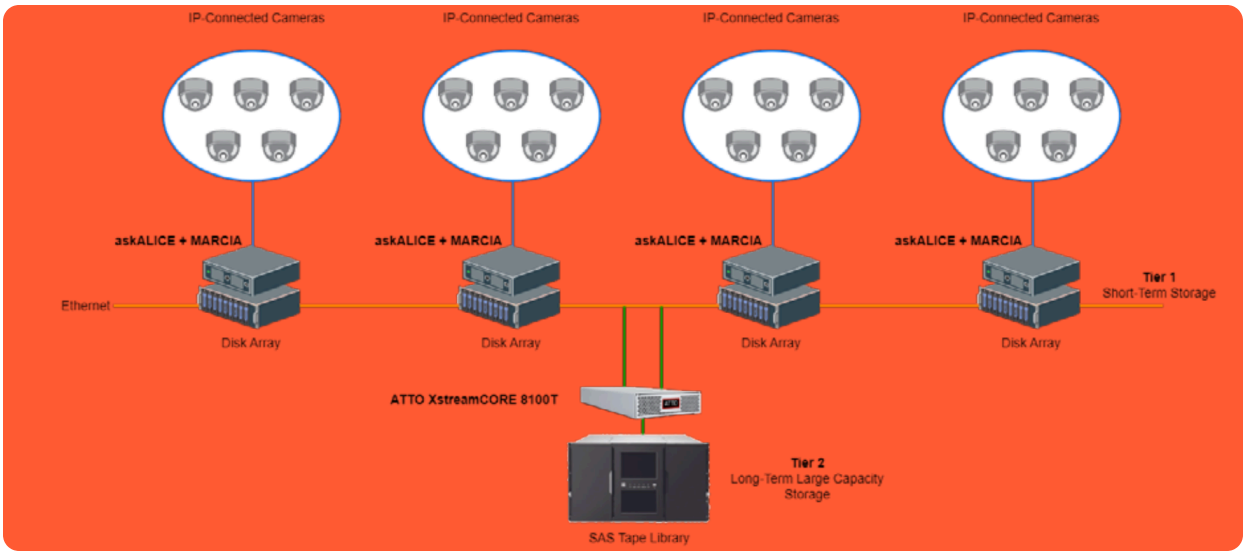
- "LTO Technology Overview," the LTO Program: https://www.lto.org/what-is-lto/
- Linear Tape-Open (LTO) Program: https://www.lto.org/
- SAS (Serial Attached SCSI) technology roadmap: https://www.scsita.org/library/serial_attached_scsi_technology_roadmap/
- Fibre Channel Industry Association (FCIA): https://fibrechannel.org/
- Microsoft documentation, Device Manager: https://docs.microsoft.com/en-us/windows-hardware/drivers/install/device-manager
- Linux SCSI Generic (sg) HOWTO: http://tldp.org/HOWTO/SCSI-Generic-HOWTO/

13. Advanced connectivity: shared access and remote LTO via Ethernet

Unlock shared LTO access with ATTO XstreamCORE

We want to highlight an important Cozaint partner, **ATTO**, whose product line integrates well into a video surveillance infrastructure and helps such environments scale.

ATTO's XstreamCORE intelligent bridges change the way organizations use direct-attached tape storage in their VMS infrastructure, by converting a direct-attached tape library or drive into a shared, Ethernet-accessible resource. The XstreamCORE 8100T lets NAS systems such as Cozaint askALICE, connected to thousands of high-definition cameras, write video to LTO tape over Ethernet using standard protocols such as iSCSI, bridging the gap between Ethernet-based storage and traditional tape infrastructure. Video moves efficiently onto high-capacity tape media, keeping it retained and available for longer.



Shared LTO over Ethernet: several askALICE + MARCIA recording systems, each with its own tier 1 disk array, share one SAS tape library through an ATTO XstreamCORE 8100T bridge

Surveillance data management and long-term retention on scalable tape

The combination of Cozaint askALICE and ATTO XstreamCORE 8100T delivers a seamless, scalable approach to video surveillance storage and retention. By extending SAS tape drive and library connectivity over Ethernet, the 8100T gives flexibility in where the tape library is physically placed. Multiple VMS servers can share and access the tape library over the network, adding architectural agility at scale.

Video captured by IP cameras is first recorded to tier 1 disk storage (NAS) for fast, real-time playback, while Cozaint's MARCIA orchestrates the flow of data to cost-effective LTO tape. The XstreamCORE 8100T extends this workflow by connecting video management systems and tape storage over the

Ethernet network. Operators keep full visibility and access through the integrated VMS timeline, so footage remains searchable and retrievable regardless of where the tape library sits on the network. This centralizes access to large-capacity LTO storage within IP-based physical security workflows, improving operational efficiency, reducing costs and keeping data within reach. The 8100T's intelligent bridging architecture accelerates data transfers, shortens backup windows and improves overall VMS storage reliability.

Key benefits

Benefit	What it means
Extend tape access across the network	XstreamCORE bridges SAS tape libraries to an existing Ethernet network, so askALICE can store video to tape from anywhere on the network, no longer limited by SAS cable length
Low-latency, high-bandwidth transfers	High-speed, uninterrupted video transfers to tape, even during active recording. With ATTO SpeedWrite™ and SpeedRead™, video from multiple tier 1 disk arrays moves efficiently to an LTO library over Ethernet
Scalable architecture	Add tape drives or libraries through additional bridges as retention grows, without redesigning the network
Centralized, shared tape storage	Multiple surveillance systems or recording endpoints can use one shared tape repository, reducing the need for direct-attached tape at every location

Ethernet-enabled vs direct-attached LTO

Feature	Ethernet-enabled SAS LTO	Direct-attached SAS LTO
Cost	$$ (lower)	$$$ (higher)
Distance	Up to 50 km with fiber	Short (up to 10 meters)
Performance	High	High
Scalability	High	Limited to direct attachment
Setup complexity	Simple to moderate	Moderate to high
Latency	Possible, but minimized by ATTO acceleration	None
Ideal use case	Distributed networks, mixed data-center networks	Small to mid-sized setups, direct-attach needs

The ATTO 8100T offers a flexible, modular approach for distributed environments, ideal for expanding tape capacity step by step as needs grow. The 8200T is designed for large, centralized tape repositories that need high-density connectivity, performance, enterprise-grade management and support for much longer distances.



ATTO XstreamCORE 8100T



ATTO XstreamCORE 8200T

ATTO XstreamCORE for SAS tape storage connectivity

Feature	XstreamCORE 8100T	XstreamCORE 8200/8200T
Network connectors	2 × 10GbE optical SFP+/RJ45 (two separate models for optical and copper)	2 × 40GbE optical QSFP+
Tape connectors	1 × 12Gb/s mini-SAS HD (4 PHYs)	4 × 12Gb/s mini-SAS HD (16 PHYs)
Network protocol	iSCSI	iSCSI, iSER
Tape devices supported	Up to 4; LTO-5 or later	Up to 16; LTO-5 or later
ATTO xCORE + eCORE	✓	✓
iSCSI CHAP and ACL support	✓	✓
Management port (RJ45)	✓	✓
Web management interface	✓	✓
SAS TLR support	✓	✓
ATTO Performance Suite	✓	✓
LTFS support	✓	✓
Power	30 W	70 W
Dimensions	1.2” H × 4.0” W × 9.4” L	1.735” H × 14.31” W × 9.90” L

Feature	XstreamCORE 8100T	XstreamCORE 8200/8200T
Weight	2.05 lb	9.7 lb
Rackmount	Up to 4 in a single 1U with ATTO FlexRack™	Yes
Warranty	3-year standard, 5-year extended	3-year standard, 5-year extended
Target use	Small to mid-size tape libraries with 1 to 4 drives	Medium to large tape libraries with 4 or more drives

Connecting an LTO tape library to Ethernet

When connecting a SAS-based LTO library to a 10GbE or 40GbE network, the ATTO XstreamCORE 8100T (10GbE) or 8200T (40GbE) acts as a protocol bridge, adding Ethernet connectivity while keeping SAS functionality.

Requirements.

- LTO tape library with a SAS interface
- ATTO XstreamCORE 8100T or 8200T bridge
- 10GbE or 40GbE network switch
- SAS cables, or a SAS fan-out cable for multiple tape drives
- 10/40GbE network cables
- 10/40GbE network interface card (NIC) in the PC server
- PC server with an iSCSI initiator

Installation steps.

1. Power off the LTO tape library and the ATTO XstreamCORE 8100T/8200T.
2. Locate the SAS ports on the LTO library and on the ATTO bridge.
3. Connect the SAS cable from the library's SAS port to an available SAS port on the XstreamCORE.
4. Make sure the connection is secure and properly seated.
5. Identify the 10/40GbE network ports on the XstreamCORE.
6. Connect a 10/40GbE cable (fiber or copper, depending on your setup) from the bridge's 10/40GbE port to a 10/40GbE switch, if multiple servers will access the tape library.
7. Make sure the network infrastructure (switches, routers) supports 10/40GbE speeds and is configured for best performance.
8. Power on the LTO library and allow it to fully initialize.
9. Power on the XstreamCORE and wait for it to boot.
10. Follow the XstreamCORE 8100T/8200T setup instructions.
11. Install a 10/40GbE NIC in the PC server, if one is not already present.

12. Install the iSCSI initiator software (built into most modern operating systems, including Windows Server, Linux and macOS).
13. Open the iSCSI initiator configuration tool on the server.
14. Add the XstreamCORE's IP address as an iSCSI target.
15. The LTO library should now appear as a storage device on the server.

14. The unbreakable chain of custody

Video surveillance systems are becoming commonplace in a society increasingly concerned with security. Their value is clear, from investigating incidents to discouraging crime. But the real test of a surveillance system's effectiveness often comes down to one crucial question: **is the video it records admissible in court?** Even the clearest video can be challenged and rejected if the integrity and chain of custody of the evidence are not properly addressed, which can make large investments in security equipment pointless.

Written from the viewpoint of a physical security and video surveillance legal specialist, this chapter examines the typical legal issues video evidence faces in federal, state and local jurisdictions across the United States, and shows how Cozaint's MARCIA, with offline LTO media tracking in the ALICE storage solutions (askALICE and ALICE1), creates a strong foundation for digital video evidence.

Why install surveillance if not for court? The foundation of admissibility

Organizations invest in video surveillance for many reasons: security, loss prevention, operational oversight, safety compliance and liability reduction. Often overlooked in the design and deployment phases is the video's potential as legal evidence. If a system cannot reliably provide footage that withstands legal scrutiny, its value in reducing risk and resolving disputes is severely diminished.

The core principle for any evidence presented in court is admissibility, which rests on three pillars:

Pillar	What it means
Relevance	The footage pertains to the case
Authenticity	The footage is what it purports to be
Integrity	The footage has not been altered

It is the last two, particularly integrity, where video surveillance faces its most rigorous legal tests.

Common legal challenges to video evidence

Opposing counsel use several approaches to challenge the admission of video recordings or seek their dismissal. Understanding them is the first step in building a robust defense for your evidence:

1. **Challenging authenticity.** Often the first line of attack. Lawyers question whether the video is the original recording and whether it accurately depicts events without alteration. They may demand proof of the recording device's reliability, its proper functioning at the time, and the identity of whoever operated or maintained it. Under Federal Rule of Evidence 901, the proponent must offer "evidence sufficient to support a finding that the item is what the proponent claims it is," through testimony from a witness with knowledge or evidence describing a process or system that produces an accurate result.

2. **Disputing the chain of custody.** Perhaps the most potent challenge to digital evidence. The chain of custody is the chronological documentation of who handled the video from the moment it was recorded until it is presented in court. Any gap, inconsistency or undocumented transfer can cast doubt on the video's integrity and lead to its exclusion.
3. **Allegations of tampering or manipulation.** With sophisticated editing software widely available, opposing counsel often allege that video was edited, manipulated or fabricated. Proving the unaltered state of digital evidence is critical; even minor unexplained discrepancies raise questions about credibility.
4. **Questioning quality and clarity.** Challenges based on low resolution, poor lighting, obscured views or digital artifacts argue that footage is too unclear to identify people or events, making it unreliable or prejudicial.
5. **Reliability of the system.** Attorneys may attack the reliability and maintenance of the entire surveillance system (cameras, network, recorders and storage), arguing it was not properly installed, calibrated or maintained.
6. **Fourth Amendment and privacy concerns.** The Fourth Amendment protects against unreasonable searches and seizures. Commercial surveillance in public areas generally carries a lower expectation of privacy, but recordings obtained by law enforcement without proper warrants in areas with a reasonable expectation of privacy can be ruled inadmissible. State wiretapping and privacy laws can also affect video with audio.

The critical role of chain of custody: best practices for digital evidence

Given these challenges, a stringent chain of custody is not merely a best practice; it is a necessity for admissibility. Procedures must be forensically sound from recording through storage, retrieval, analysis and presentation.

Documentation, documentation, documentation. Every step of the evidence lifecycle must be documented:

Stage	What to record
Collection	Who collected the evidence, when (date and time), where, and the method used
Handling and access	Every person who accessed or handled the video: name, badge number, date, time and purpose
Storage	How and where the evidence was stored (for example, a secure server or encrypted LTO tape), including physical and digital security measures
Transfer	When and how the evidence moved between authorized people or locations
Analysis	Any analysis performed, by whom, and with what tools

- **Secure storage environments.** Prevent unauthorized access, tampering and environmental damage, with physical security (restricted access to servers and tapes) and digital security (encryption, access controls, audit logs).
- **Forensic copies.** Never analyze or manipulate the original. Create forensic copies with write-blocking tools and perform all analysis on the copies.
- **Integrity verification (hashing).** Apply cryptographic hashing (for example, SHA-256) at key stages: on recording, on transfer to long-term storage and before analysis. A matching hash shows the data has not been altered or corrupted; any change is an immediate red flag.
- **System reliability documentation.** Keep records of maintenance, calibration, software updates and hardware checks to show the system was functioning reliably.

Cozaint MARCIA: the foundation of reliable video storage

MARCIA addresses long-term video retention and reliable access, the first critical layer of legally sound video evidence. Disk-only solutions become prohibitively expensive for long retention, forcing organizations to discard valuable footage early. MARCIA's multi-tier approach combines a first tier of RAID-protected disk (NAS) for immediate access to recent footage with a second tier of LTO libraries for scalable, cost-effective long-term archive. The benefits for legal evidence:

- **Extended retention.** High-resolution video can be kept for months or years, so footage of incidents discovered much later is still available, which is vital for a complete timeline and chain of custody.
- **High-resolution evidence.** Long-term storage of high-resolution video directly counters challenges about quality and clarity.
- **Structured archiving.** A systematic, multi-tier approach is a more controlled and secure way of handling data than ad hoc solutions, supporting arguments for integrity.

Even with strong storage, it remains necessary to demonstrate the chain of custody and the video's undamaged condition after it leaves the recording environment and moves to archival media such as LTO.

MARCIA and the LTO Cartridge Memory chip: validating the chain of custody

Every LTO cartridge contains an embedded **Cartridge Memory (CM) chip**: a non-volatile, contactless chip that stores metadata about the tape, including:

CM-chip data	Why it matters
Manufacturer name and serial number	Unique identifiers for the physical cartridge
Tape mount count	The number of times the cartridge has been loaded into a drive. This count is key to determining chain of custody

CM-chip data	Why it matters
Data written and read	The amount of data transferred to and from the tape
Diagnostic and log information	Records of drive activity and any errors

Cozaint uses this technology to provide a high level of chain-of-custody validation. As LTO cartridges holding surveillance video are written and archived by MARCIA, the ability to read and track the CM-chip data is critical:

1. **Baseline recording.** When a new cartridge enters the MARCIA system, its initial CM-chip data is recorded, including its tape mount count (ideally zero or one after initial formatting).
2. **Continuous monitoring.** As the cartridge is mounted by the ALICE system to write video, MARCIA logs each interaction, creating a digital audit trail that mirrors the physical chain of custody.
3. **Integrity validation.** If a cartridge holding crucial evidence is later presented in court, an expert witness can use MARCIA's records to read the current CM-chip data and compare the tape mount count and other metadata with the documented history. Any discrepancy becomes apparent:
    - **Unauthorized access:** a mount count higher than the documented handling shows the cartridge was loaded at an undocumented time, a red flag for possible unauthorized access or tampering.
    - **Alteration detection:** the CM chip does not store the video itself, but unexpected changes in data written or diagnostic logs, together with unauthorized mounts, can show the integrity of the video may have been compromised.

This provides a strong evidentiary link to the physical handling of the media, which is directly tied to the video's integrity.

A strong defense for video evidence

- **Authenticity:** robust storage, systematic archiving and the traceable nature of LTO media support the argument that the video came from a reliable source and has been preserved as recorded.
- **Integrity:** chain-of-custody tracking through CM-chip data, combined with secure storage, makes tampering allegations much harder to sustain. An attempt to alter a cartridge would be expected to change its CM-chip data, which serves as both a deterrent and a detection mechanism.

Beyond paper logs or even digital hashes of the video clip alone, this chip-level tracking adds a strong layer of authentication: a concrete, embedded record of the physical cartridge's history. Together, MARCIA's affordable long-term retention of high-resolution video and its offline LTO media tracking address the core concerns of authenticity and integrity, and give expert witnesses technical evidence that the media holding the video was handled according to strict protocols.

The role of the expert witness

In any legal challenge involving complex digital evidence, the expert witness is essential. An expert familiar with Cozaint's MARCIA can:

- **Educate the court:** explain how the video was recorded and stored, and how the LTO CM chip and MARCIA track the chain of custody.
- **Testify to system reliability:** describe the design, implementation and maintenance of the MARCIA system.
- **Validate the chain of custody:** present MARCIA's audit logs together with the CM-chip data, explaining how the tape mount count corroborates documented handling or flags possible tampering.
- **Counter opposing arguments:** rebut challenges to authenticity and integrity with a clear technical explanation of the safeguards in place.

Legal precedents and the future of video evidence

U.S. courts, guided by the Federal Rules of Evidence and state laws, consistently require strong foundational evidence for digital media. Cases such as *United States v. Browne* (2009) and *United States v. Jackson* (2009) highlight the importance of proper authentication of digital evidence. Specific rulings on LTO CM-chip validation are still evolving, but the principle aligns with the growing demand for verifiable digital chain of custody. *State v. Casey Anthony* (2011), while not involving video, showed how an unclear chain of custody for digital evidence (in that case, internet search histories) can weaken a case.

Privacy remains a critical consideration. Commercial surveillance in public spaces is generally permissible, but organizations must be mindful of privacy expectations in less public areas.

Building legal admissibility into system design is becoming a requirement rather than an option in physical security and video surveillance. A successful prosecution or defense can depend on the ability to defend video evidence under legal scrutiny. MARCIA provides the framework for long-term, high-resolution video storage, so footage is available when needed, and elevates chain-of-custody validation through tracking of LTO CM-chip data: a chip-level record of physical media handling that is a strong defense against tampering and chain-of-custody challenges.

For any organization concerned about the legal value of its surveillance video, the sensible, forensically sound course is to invest in solutions that preserve an uninterrupted chain of integrity from the first recording to the presentation in court.

This chapter is general information, not legal advice. Consult counsel about evidence requirements in your jurisdiction. # 15. Best practices and getting started

Design

1. **Record at the quality the cameras were bought for.** Change the storage, not the evidence.
2. **Size the disk tier for how far back operators usually look,** typically days to a few weeks, about 10% of total retention as a starting point.
3. **Put the full retention period on LTO,** with room to grow by adding cartridges.
4. **Use at least two tape drives,** so recalls and recording never compete.
5. **Use RAID on the disk tier** to protect recent video and keep performance high.
6. **Avoid SMR hard drives** in recording systems.
7. **Decide your protection policy:** offline cartridges for air-gapped copies, WORM media where recordings must not be overwritten, and encryption where needed.

Operation

- **Keep operators in the VMS.** The test of the design is whether they can ignore where the video is stored.
- **Set expectations:** recent video is instant; older video on tape typically returns within minutes.
- **Monitor the system:** library health, drive cleaning and calibration, cartridge use and retention policy.
- **Train staff** on playback, export and evidence handling.

Getting started

1. **Start with your numbers.** Use the storage calculator at cozaint.com/assessment to size your system and compare MARCIA, disk and cloud, and download the PDF report.
2. **Evaluate what you have.** Review current capacity, retention, playback experience and cost, including power and replacement.
3. **Start small if you like.** A pilot on one site or one group of cameras proves the workflow with minimal risk.
4. **Work with an experienced partner.** Cozaint and its reseller partners design, install and support lifecycle architectures for enterprise surveillance.

The benefits, in summary

Benefit	What it means
Cost-effective	Substantially lower lifecycle cost than disk-only or cloud, growing with scale
Scalable	Grow retention by adding cartridges; grow libraries by adding modules

Benefit	What it means
Fast where it matters	Recent video instantly from disk; older video within minutes
High capacity	30 TB per LTO-10 cartridge, up to 40 TB with the newest cartridges, and a roadmap to 576 TB
Long-lived	Media rated for decades in archive
Energy efficient	No power for cartridges at rest
Secure	Offline copies, WORM media and encryption
Operator-friendly	The patented MARCIA workflow keeps every recording on the same VMS timeline

Media & Entertainment proved LTO for video. MARCIA makes it practical for surveillance. Patented VMS integration makes the storage tier invisible to the operator.

Talk to Cozaint. Request a MARCIA assessment at **cozaint.com/marcia**, run your numbers at **cozaint.com/assessment**, or contact **info@cozaint.com** · **760-975-8000**.

About the author and Cozaint

Jay Jason Bartlett is the CEO of Cozaint Corporation and the author of the first edition of this book (2023).

Founded in 2018, Cozaint Corporation is a service-disabled veteran-owned business that develops video surveillance and storage solutions, including MARCIA™ Intelligent Video Lifecycle Management, askALICE™, ALICE1™ and BOBBYvms. Cozaint is headquartered in Palmetto Bay, Florida, with an office in San Diego, California.

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